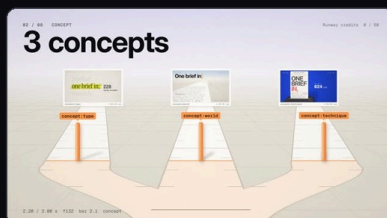
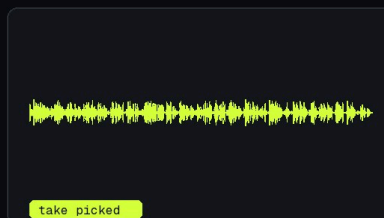


One brief in, *a finished* motion video out.

How the motion-studio pipeline works, and how to make a film with it: from the first sentence of your brief to masters in every format.



1 Concept, pitched with frames
research, a concept, key frames in every format



2 You hear it and pick
song, narrator or music bed: two takes each



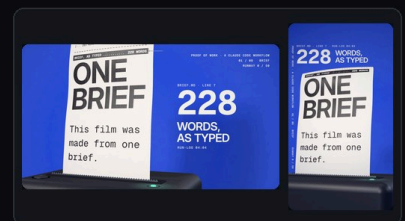
3 Every shot on the beat
timed from the audio, never by hand



4 Styleframes, then a critic
the standard every shot is built to



5 Checked by tools and eyes
cuts, pace, text, loudness; an Opus critic



6 Delivered in every format
16:9, 9:16, 1:1, 4:5, 4K, mastered sound

OPEN SOURCE · MIT

github.com/winchxyz/motion-studio

01 · WHAT IT IS

A motion designer you brief in one paragraph

motion-studio is a Claude Code workflow with its own animation engine. You describe a video; agents research the subject, pitch a concept, make the sound, build every shot in code, check the result and deliver masters in every format. You make one decision along the way: the look and the sound.

one decision**You pick, it makes**

The run stops once: you see the concept's key frames and hear the audio takes, choose, and it runs to delivery.

every frame is code**Nothing is a template**

Each frame is drawn on the GPU in headless Chrome and encoded with ffmpeg: type, shapes, 3D, generated art, all animated in code.

checked twice**Tools and an outside eye**

Automatic checks for cuts, pace, text and loudness, then a critic agent that judges the film from its frames only.

What it makes

- Brand and product spots, launch films, explainers
- Music videos with lyrics timed word by word
- Narrated films over a music bed
- Social cut-downs and showreels
- 16:9, 9:16, 1:1, 4:5 and 4K from one timeline, each format laid out on its own, with the platforms' safe areas

What you need

- [Claude Code](#)
- Node 22+, Python 3.11+, ffmpeg on the PATH
- Google Chrome (or `CHROME` pointing at a Chromium)
- A GPU helps: every frame is rendered on it
- Optional: the Runway connector (MCP) for generated images, music and voice. Without it, music is synthesized and art is drawn in code.

Who does what

Claude Code runs the workflow `.claude/workflows/motion-video.js`. The agents that make things run on Claude Sonnet 5.5; the critic that judges the film runs on Claude Opus 5.5. Underneath sit the studio's engine (`studio/engine`), its tools (`studio/tools` : render, QA, audit, delivery) and its skills (`.claude/skills`): the written method for each step and the quality bar every agent reads.

02 · THE PIPELINE

Six steps, one of them yours

By default the workflow runs **direct**: after your decision, one agent makes the whole film the way a single motion designer would. For a short film that is faster and cheaper than splitting the work.

- | | | |
|----------|--|-----------------------------|
| 1 | Setup
The film's folder from a template, its length and formats, a short brief written from your words. | 1 agent
about 5 min |
| <hr/> | | |
| 2 | Concept and sound, at the same time
One agent researches the facts and draws the concept as key frames in every format. Another makes two audio takes (a song, a music bed, or narrator voices) and the timing every later step reads. | 2 agents
about 25-30 min |
| <hr/> | | |
| 3 | Your pick
You look at the key frames, listen to the takes and choose. Or ask for other takes, or a different direction. Nothing is built on audio you have not heard. | you
as long as you like |
| <hr/> | | |
| 4 | Make
One agent writes the shot list, sets the style and builds every shot, checking as it goes with contact sheets, <code>qa.py</code> and the text audit. | 1 agent
about 45-75 min |
| <hr/> | | |
| 5 | Critic
A critic on Opus that has never seen the code judges the whole film from its frames: the hook, pacing, type, composition, art direction. | 1 agent
about 5-10 min |
| <hr/> | | |
| 6 | Finish
Fixes the critic's major notes, renders every format, masters the sound to -14 LUFS, makes the phone preview, posters, styleframes and post copy. | 1 agent
about 15-30 min |

The times are estimates for a 15-30 second film on a laptop GPU. Measure any run with python `studio/tools/runcost.py <run id>`: agents, tokens, dollars at API prices and hours per step.

Where it runs

Everything happens in your Claude Code session and on your machine. You can watch a run live with `/workflows`, and Claude tells you when it needs you and when the film is delivered.

03 · PRESETS

Pick the size of the crew

The same workflow runs three ways. Ask for one by name; direct is the default.

PRESET	AGENTS	HOW IT WORKS	USE IT FOR
direct	6	Setup; concept and sound at the same time; your pick; one maker; one critic; one finishing pass.	Films up to about a minute. The default.
lean	about 12	A planner writes a shot table and a note per shot; a look stage builds a shared kit and two hero shots; builders work on groups of shots in parallel; one review round.	Long films with many shots, where parallel builders save time.
full	30+	Three concepts with prototypes and a judge panel; a drawn storyboard and animatic with a critic; a builder, reviewer and fixer for every shot; up to three review rounds.	When a film is worth everything.

What time and cost come from

Every agent starts by loading the studio and the film, and every step it takes re-reads all it has seen so far. Nearly all of the usage is that re-reading, so a film costs less when fewer agents read less:

- **Fewer agents.** Direct uses six; full uses thirty or more.
- **Short documents.** brief.md 6 KB, concept.md 5 KB, the shot table 12 KB, a note per shot 1.5 KB. Research and evidence live in their own folders, which the makers do not read.
- **Budgets.** About three render-and-look passes per task, test renders at two motion-blur samples, masters at eight.
- **The right model.** Makers on Sonnet at medium effort, only the critics on Opus.

Measured: the first long film (48 seconds) ran the full path with 46 agents, about 17 hours and about \$320 of usage at API prices. That is why direct became the default. On a subscription the same usage counts against your weekly limit instead of a bill.

Runway credits

Generated images, music and voice cost Runway credits. The workflow caps them per run (`credits` , default 150), says what a batch costs before running it, and logs every generation with its cost in `assets/gen/manifest.json` . Typical: two music takes 8-16, two narrator voices about 12 each.

04 · GETTING STARTED

From clone to first film

1. Install

```
git clone https://github.com/winchxyz/motion-studio
cd motion-studio
npm install
pip install -r requirements.txt
```

Check that ffmpeg and Chrome are found: `ffmpeg -version`, and the engine test renders with `node studio/tools/render.mjs _lab/engine-test --sheet 0:8:1 --cols 8 --tw 240`.

2. Open Claude Code in the folder and brief it

YOU, IN CLAUDE CODE

Run the motion-video workflow: a 30-second launch film for Acme Notes, the note app for field teams. Show the offline sync and the photo notes. Their brand: `acme.example/brand`. 16:9 and 9:16. A music bed, no narrator.

Claude starts the workflow and tells you what it will do and roughly what it costs. Setup, the concept and the sound take about half an hour; then it comes back with key frames and audio takes for you.

3. Pick, then let it run

Answer in plain words, "Take B", and it makes the film, has it critiqued, fixes it and delivers. When it is done you get the phone preview in the chat and the masters in the film's `out/` folder.

Briefs that work well

product spot

"A 20-second spot for our budgeting app: one problem, the app solving it in three screens, the App Store line. 9:16 for Reels and TikTok, upbeat music bed."

music video

"A 45-second lyric video for our track (attached), big kinetic type at the chorus, subtitle-size in the verses, riso print look, 16:9 and 9:16."

explainer

"A 40-second explainer of our API in four steps, narrated by a calm male voice, clean diagrams, our colours, 16:9, ending on the docs URL."

05 · THE BRIEF

What a good brief says

The brief is the only thing the workflow knows about your film before your pick. A few plain sentences beat a long one: say what matters and leave the craft to the agents.

SAY	WHY IT MATTERS
What it is	In plain words, the way you would tell a friend. The film will say it on screen.
Who it is for, where	X, TikTok, a launch page, a talk: it sets the pace, the formats and the hook.
The one message	One idea per film. Everything else supports it.
The facts, with sources	Numbers, names and handles go on screen only after they are checked at their source. Link the source.
Length and formats	15, 30 or 45 seconds; 16:9, 9:16, 1:1, 4:5, 4K.
The look	A style card from <code>styles/README.md</code> , a reference link, or three adjectives. Name the brand kit if there is one.
The sound	A song, a music bed, a narrator (male or female, the tone), or your own track.
Must and must-not	What has to be shown, what must never be said (prices, promises, competitors).

weaker

"Make a cool video about our app that goes viral."

No message, no facts, no length: the agents have to guess what matters, and the film says little about the app.

stronger

"30 s for X: what our app is (a shared grocery list for families), how to start (free on iOS and Android), the one number we are proud of (with its source), our colours and our humour. 16:9 and 9:16, a music bed."

The film can say what it is and how to start, in the brand's own voice.

06 · THE STOP

Look, listen, pick

About half an hour in, the run stops and Claude sends you two things. This is the cheapest moment to change anything: nothing has been built yet.

the look

Key frames

The concept's hook and one key moment, drawn by the engine in every format you asked for. They are in `<film>/concept/out/stills/`, and the concept itself is in `concept.md`.

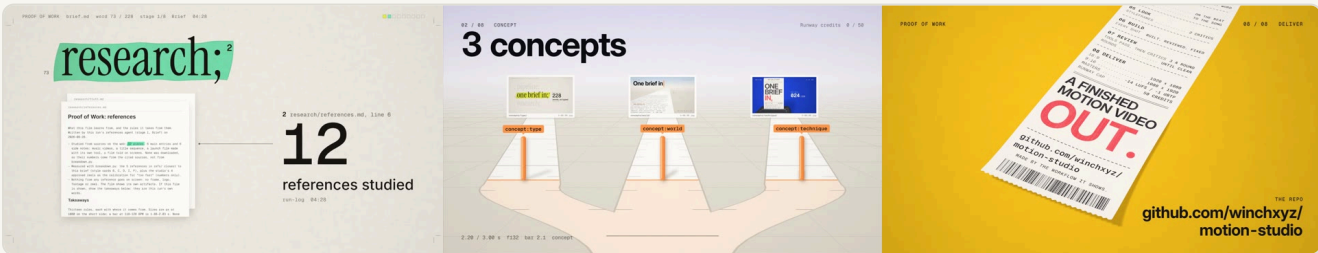
the sound

Takes

Two takes of the song or the music bed, or each narrator voice over each bed, cut to the length the film will use. They are in `<film>/out/listen-* .mp3`.

How to answer

YOU SAY	WHAT HAPPENS
"Take B"	It sets that take in film.json and continues to delivery. No more stops.
"Voice 2, bed A, music quieter"	The voice and bed you chose, with the mix changed before anything is built.
"Neither: slower, warmer, no vocals"	Two new takes (a few more Runway credits, said before it spends them), then you pick again.
"No song, a male narrator"	The sound step runs again with a narration script and two voices over a music bed.
"Keep the idea, bigger product"	The note goes to the maker with the approved concept.
"Different direction: more photographic"	The concept step runs again with your note before anything is built.



Concepts are pitched with frames the engine draws, not with descriptions.

07 · DELIVERY AND NOTES

What you get, and how to change it

The files

FILE, IN <FILM>/OUT/	WHAT IT IS
<name>—16x9-master.mp4	The master: full quality, sound mastered to -14 LUFS and -1 dBTP. One per format (9×16, 1×1, 4×5, 4k).
<name>—16x9.mp4	The share file: smaller, for posting.
<name>—16x9-preview.mp4	The phone preview, under 30 MB; Claude sends it to you in the chat.
<name>—16x9-poster.png	A poster frame for thumbnails and link cards.
<name>—16x9-styleframes.jpg	A sheet of the film's key frames.
<film>/post.md	Post copy for X and Reddit, leading with the hook.

Giving notes

Open the preview player: run `node studio/tools/serve.mjs` and go to `127.0.0.1:8960/studio/engine/preview.html?film=<film>`. Scrub to a moment, press **N** and type the note: it lands in `<film>/notes.md` with its time and format. Or simply tell Claude what to change.

Then ask Claude to run the film again from the make step with your notes. The maker reads them, continues from the shots that are already built, and the critic and the finishing pass follow as before.

If a run is interrupted

A dropped connection, a usage limit, a closed laptop: finished work stays in the film's folder. Ask Claude to resume the film from the step it reached (`from` and `film`); it continues instead of starting over.



The preview player: real-time playback, beat and edit marks on the timeline, formats, safe areas, notes.

08 · OPTIONS

Everything you can ask for

You rarely need these: say them in plain words and Claude passes them to the workflow.

OPTION	WHAT IT DOES	DEFAULT
brief	Your words, passed on verbatim.	required
preset	<code>direct</code> , <code>lean</code> or <code>full</code> : the size of the crew.	direct
formats	<code>h 16:9</code> , <code>v 9:16</code> , <code>s 1:1</code> , <code>p 4:5</code> , <code>k 4K</code> .	h and v
refs	Links or files to learn from; they are downloaded and broken down (pace, cuts, palette, tempo).	none
credits	The most Runway credits the run may spend.	150
film + from	Resume a film at a step: brief, concept, sound, plan, look, build, review, deliver.	start
until	Stop after a step, for example after the plan.	deliver
notes	Directions for every agent in the run: your notes on a cut, things to keep, things to avoid.	none
gate	<code>look</code> also stops after the hero shots (lean and full).	none
rounds	Review rounds in lean and full.	lean 1, full 3
samples	Motion-blur samples per frame in the masters.	8
models	<code>{ worker, judge, builderEffort }</code> : who makes and who judges.	Sonnet makes, Opus judges
listen	<code>false</code> skips your pick. Not recommended.	true

09 · TOOLS

The tools under the workflow

The workflow uses the same commands you can run yourself, from the studio root.

```
# a new film from a template
node studio/tools/new-film.mjs my-film --template spot          # or music-video
# the preview player
node studio/tools/serve.mjs                                # 127.0.0.1:8960/studio/engine/preview.html?
film=/my-film
# look at the film: a contact sheet, or full-size stills
node studio/tools/render.mjs my-film --sheet 0:30:0.5 --cols 8 --tw 320
node studio/tools/render.mjs my-film --stills 1.5,7.2
# render a format, then master and package everything
node studio/tools/render.mjs my-film --video --format h
node studio/tools/deliver.mjs my-film
# checks: cuts and pace, then every piece of text
python studio/tools/qa.py my-film --format h
node studio/tools/render.mjs my-film --audit 0:30:0.1 && node studio/tools/audit.mjs my-film
# a song: beats, bars and lyrics timed word by word
python studio/tools/song.py my-film/assets/song.mp3 --lyrics my-film/lyrics.txt
# study a reference video: sheets, cuts, pace, palette, tempo
python studio/tools/breakdown.py refs/clip.mp4
# a font from Google Fonts (OFL) into the film
node studio/tools/fonts.mjs "Space Grotesk" my-film
# what a workflow run cost: agents, tokens, dollars, hours per step
python studio/tools/runcost.py <run id>
```

What the engine draws

Layers of shaders, Canvas 2D, images on a bendable mesh, image sequences from generated video, Three.js scenes with extruded type and logos; looks such as halftone, riso, dither, ink, kuwahara, CRT and grain; kinetic type, typewriter, odometer and lyric captions; beat grids from a BPM or a song; 11 transitions; motion blur and bloom. Every frame is a pure function of time, so any frame can be rendered alone and checked.

10 · THE BAR

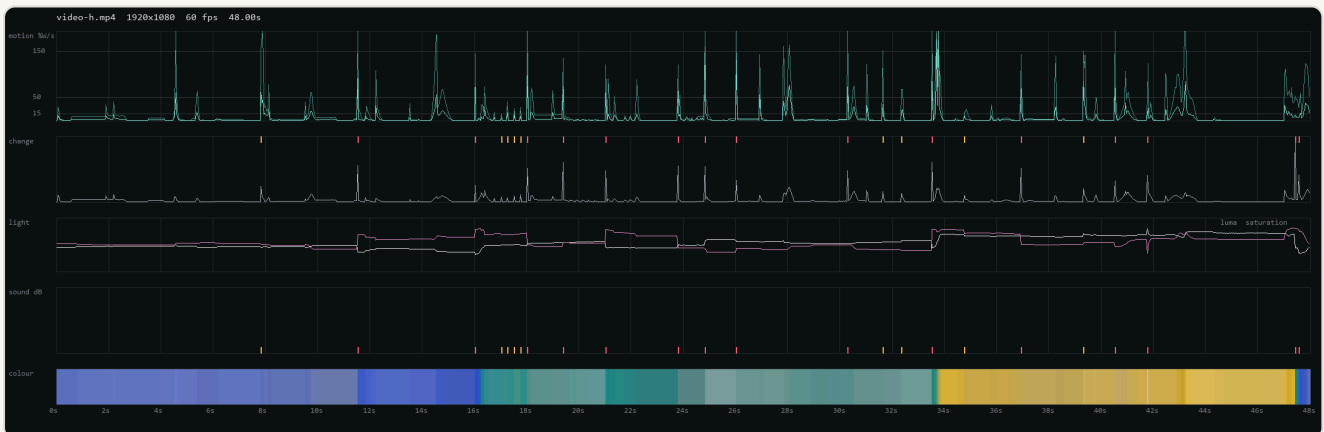
What every film has to pass

The rules the agents work to

- **One idea at a time.** Every line of text stays readable at least 1 s, 0.25 s per word.
- **Motion snaps, then holds.** Moves of 0.2–0.5 s, then a readable rest; nothing faster than a whip outside a transition.
- **Safe areas per format,** so platform buttons never cover the text.
- **The truth on screen.** Every fact, number and handle checked at its source and written in the brief.
- **Official brand assets only;** never a redrawn logo or mascot.
- **Generated media is a base,** redrawn and animated in code, never pasted in as is.
- **No placeholders,** ever, in anything you are shown.

The checks that enforce them

- **qa.py** : cuts nobody placed, stretches too fast to follow (optical flow), light, colour.
- **audit.mjs** : every piece of text for safe area, size, overlaps and reading time, in every format.
- **Loudness:** -14 LUFS integrated, -1 dBTP peak on every master.
- **The critic:** an Opus agent that sees only the frames, never the code.
- **Your eyes:** the key frames and takes before anything is built, the preview after.



qa.py on a finished cut: motion, scene change, light, sound and colour across the whole film. Every cut it finds must be one the plan declared.

11 · WHEN SOMETHING GOES WRONG

Troubleshooting

WHAT YOU SEE	WHAT TO DO
Runway tools are missing	Reconnect the Runway connector in Claude. Until then, music is synthesized and art is drawn in code.
A run stopped halfway	A dropped connection or a usage limit. Ask Claude to resume the film from the step it reached; finished work is kept.
A step takes much longer than its estimate	An agent is usually polishing. Ask Claude to check its last steps; it can stop it and continue from the next step.
The delivered sound is not your take	<code>film.json</code> 's <code>audio.song</code> must point at the take you picked; delivery mixes it with the effects.
Text is cut by the platform's buttons	Run the text audit for that format; the safe areas come from <code>studio/engine/formats.js</code> .
The film feels too fast	The most common note on a first cut. Say so; the maker lengthens holds and removes ideas before it speeds anything up.

Links

Repository	github.com/winchxyz/motion-studio
The workflow, step by step	WORKFLOW.md
Overview and quick start	README
This guide	docs/motion-studio-guide.pdf
Claude Code	claude.com/claude-code

Built by [@winchxyz](#) with Claude Code. Claude Opus 5.5 wrote the engine, the tools and the workflow; when it runs, Sonnet 5.5 makes and Opus 5.5 judges. Code under the MIT License, fonts under the SIL Open Font License. Issues and pull requests are welcome.